

Zarif Safwan Hoque

4th Year
Computer Science and Engineering
Islamic University of Technology

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PROFESSIONAL SUMMARY

Motivated Associate Software Engineer with a strong foundation in Java and Spring Framework, complemented by experience in full-stack web development. Skilled in Agile methodologies, collaborative teamwork, and delivering high-quality, maintainable code. Quick learner with excellent problem-solving abilities and a keen attention to detail. Eager to contribute technical expertise and grow within a dynamic software development environment.

EDUCATION

•Islamic University of Technology

Bachelor's in Computer Science and Engineering

CGPA: 3.84

•Willes Little Flower School and College

A Level

Result: 4 A*

•Willes Little Flower School and College

O Level

Result: 7 A* 1 A

WORK EXPERIENCE

•Chaldal.com

Engineering Intern

May 2024 – July 2024

- Worked with company production tech stack, which included technologies such as F#, React and Fable.
- Contributed to the backend system including bug fixes.
- Applied agile development workflows and collaborated with engineering mentors on sprint tasks.

PROMINENT PROJECTS

Topic Modeling for Bengali Text using BERTopic

Python

- Built an end-to-end NLP pipeline for Bengali Wikipedia (**70k+ articles, 18M+ words**) using **BERTopic**.
- Engineered a custom preprocessing + embedding + clustering workflow with UMAP, HDBSCAN, and C-TF-IDF to generate highly interpretable topics.
- Developed an interactive **Streamlit web app** for real-time topic exploration and automatic classification of new Bengali documents.
- Demonstrated the application of transformer-based topic modeling for Bengali, opening possibilities for content recommendation and trend analysis

SpectraNet: Lightweight Hybrid Deep Learning for Cloud Workload Prediction

Python, PyTorch, LSTM, CNN, SENet, Frequency-Domain Analysis

- Developed **SpectraNet**, a hybrid time-series forecasting model combining LSTM, CNN, and Squeeze-and-Excitation networks for accurate CPU usage prediction.
- Incorporated both **time-domain and frequency-domain features**, capturing short-term patterns and long-term dependencies in cloud workloads.
- Evaluated on large-scale datasets (**Alibaba Cluster Trace 2018, VMCloud**), achieving significant gains in accuracy and efficiency over existing baselines.
- Demonstrated practical value for **cloud resource scheduling and autoscaling**, reducing cost and improving system reliability.

HealthTrack

React Native, Firebase

- Designed **HealthTrack**, a cross-platform **mobile application** using React Native, with Firebase for real-time **backend integration** and data storage.
- Developed features for setting fitness goals, tracking daily progress, and visualizing user performance to enhance engagement and retention.
- Ensured responsive UI/UX and secure **data handling**, leveraging Firebase Authentication and Firestore.

ShopSmart

HTML, CSS, JavaScript

- Developed **ShopSmart**, a dynamic **ecommerce web application** using HTML, CSS, and JavaScript with a focus on responsive design and smooth user experience.
- Developed core features including location-based search, anonymous user interaction, and standard buyer-seller workflows such as product listing, cart management, and checkout.
- Optimized frontend usability with form validation, modular layout, and optimized page load performance.

Zibra

JavaFX, MySQL

- Engineered **Zibra**, a JavaFX-based desktop application enabling anonymous blog sharing and discussions.
- Developed a **secure backend** using MySQL for managing user-generated content with data integrity and access control.
- Developed a dynamic and intuitive UI with JavaFX to support smooth navigation, post creation, and real-time discussion features.

PoorCAD

SFML, C++, C

- Designed and built **PoorCAD**, a lightweight drawing application using SFML, applying **Object-Oriented Programming (OOP)** principles.
- Applied real-world OOP concepts such as encapsulation, inheritance, and polymorphism to manage interactive drawing features.

TECHNICAL SKILLS

Languages: C, C++, Java, Python, JavaScript

Frameworks: JavaFX, Java Spring Boot, React Native, PyTorch

Web Technologies: HTML, CSS, JavaScript

Databases: MySQL, Oracle, Firebase

Version Control: Git

Operating Systems: Linux, Windows

Relevant coursework: Structured Programming, Discrete Mathematics, Digital Logic Design, Data Structures, Database Management Systems, Object Oriented Programming, Algorithms, Visual Programming, System Analysis and Design, Operating Systems, Software Engineering and OOD, Web Programming, Mobile Application Development, Machine Learning

Tools & Platforms: Docker, Postman, Figma, VS Code, IntelliJ IDEA, Android Studio

DevOps & CI/CD: GitHub Actions, Jenkins (basic), Agile Development, Scrum

Cloud Services: Firebase, Heroku, AWS (basic)

EXTRACURRICULAR ACTIVITIES

1. Active member in IUTCS society in event management and participation
2. Active participant in online competitive programming platforms
3. Former member of the Bangladesh Scouts Organization and achieved President's Scouts award
4. Achieved World highest medal in O level Maths and Daily Star awards and Edexcel High Achiever's Award